



Agilent's RF circuit design teaching solution

the coming time we will surely come out with that," says Kapil Kumar Garg, co-founder and CEO, ThinkWare—a venture of KC Robotics & Embedded.

"Various types of educational kits covering the whole spectrum of engineering education—especially with PC interfacing, data logging and automation—are in demand by institutes and colleges," adds Ilesh Shah, proprietor, Abvolt Technologies.

Some educational kit providers are coming up with kits in sync with the latest trends in technology.

"Kits'n'Spares is providing kits on latest boards like Arduino, and is also promoting Raspberry Pi through its partnership with Element14. Plus, it has recently launched USB-based boards for 8051 and PIC, and will soon be launching a board for the AVR chips," informs Neena Chopra, director, Kits'n'Spares.

"Development boards like the Raspberry Pi and Arduino have created large communities of engineers and entrepreneurs around them. While some are creating clones of these boards and modules and targeting the hobbyist community, the more enterprising ones have gone ahead and created industry-ready solutions using these boards. Hence we are seeing an interesting trend where a cottage industry of sorts is developing around what were perceived to be educational kits," she adds.

In robotic kits, programmable ro-

bots are the most popular. These kits come with sensors, motors and a robot brain. One can connect the sensors to the brain according to one's application and program the brain to perform different kinds of behaviours. These kits are available for schools as well as colleges with a proportional level of complexity.

Each robotic kit features either pre-assembled or unassembled PC boards, hardware and mechanical drive. Only basic hand tools are required for assembly. These fascinating robotic kits allow one to experience and learn any of the features like sound sensor, remote control, infrared sensor, wired control and/or programmable memory.

Major consumers and key drivers

Major consumers of educational kits are school students, engineering students, hobbyists and R&D houses of electronics companies. Accessibility, step-by-step guidance and application notes are the key drivers for these kits.

Educational kits have become a

Key drivers of educational kits in various verticals

- **Students/teachers/academia.** The key driver for these kits in education is employability and the industry relevance. Professional-grade lab solutions are most preferred for the purpose.
- **Hobbyists/independent researchers.** For hobby projects and independent research, relevance to the subject is important. Some retailers and specialised channel partners have jumped in to serve this need.
- **Government/NGOs.** The key driver for these kits is their simplicity, ease-of-use and interesting user interface. NGOs and government bodies deploy these kits to generate interest in a particular stream of science and research.

— V.V. Pathy Iyer, business development manager—Education, Electronics Measurement Group, Agilent Technologies India

quintessential part of labs and curriculum of every professional education programme in our country. These kits are the students' first sneak peek into the practical aspects of their subjects and their industrial or professional relevance in life. Educational kits are also being deployed early at the school level to instill interest in various exciting streams of science and biology.

India is strategically looking at advancing its already established position of developing great engineers and scientists for the world. These educational kits have a small but significant contribution towards it. With public-private partnerships, special programmes are being deployed to generate interest in science and research early in life and educational kits are a key tool for the same. The same is being copied at the elementary school level.

“Earlier, lab kits were very costly. Now the cost has gone down steeply. Low-cost 8-/32-bit microcontroller kits are available. Future belongs to targeted DIY kits that are designed to solve particular problems —Kapil Kumar Garg, co-founder and CEO, ThinkWare